

Handling Quorum Issues in MySQL InnoDB Cluster

Issue Description

When a MySQL InnoDB Cluster loses quorum, it is unable to process write transactions. The cluster enters a read-only state and displays the following error when viewed from an affected node:

```
"status": "NO_QUORUM",  
"statusText": "Cluster has no quorum as visible from 'mysql2:3306'  
and cannot process write transactions. 2 members are not active."
```

This issue can occur due to network failures, node crashes, or ungraceful shutdowns of multiple members.

Resolution Methods

Method 1: Reboot the Complete MySQL InnoDB Cluster

Use this method when the entire cluster has gone down or is unreachable and needs to be restarted from a major outage.

Steps:

1. Stop MySQL Services on all cluster instances.
2. Start MySQL Services on all instances, one by one.
3. Once MySQL services are running on all nodes, connect to any node using the MySQL Shell:

```
# mysqlsh
```

4. Reboot the cluster from the complete outage:

```
# var cluster = dba.rebootClusterFromCompleteOutage()
```

5. After a successful reboot, check the cluster status:

```
# var cluster = dba.getCluster();  
# cluster.status()
```

✓ Expected Result: The cluster status should show "OK", indicating that quorum has been restored and all nodes are participating.

Method 2: Force Quorum on a Single Node

-  Use with caution. Only proceed if:
 - You are certain the chosen node has the most recent and consistent data.
 - The other nodes are offline or unrecoverable.
 - Steps:
6. Access the MySQL Shell on the chosen primary node:

```
# mysqlsh
```

7. Get the cluster object and force quorum:

```
# var cluster = dba.getCluster();  
# cluster.forceQuorumUsingPartitionOf('clusteradmin@mysql2:3306')
```

8. Confirm the cluster status and verify that the selected node is in Read/Write mode:

```
# cluster.status()
```

9. Rescan the cluster to detect and reintegrate any other nodes:

```
# cluster.rescan()
```

10. If additional nodes are detected and eligible to rejoin, use the following commands:

```
# cluster.rejoinInstance('mysql1:3306');  
# cluster.rejoinInstance('mysql3:3306');
```

 Expected Result: The cluster should restore quorum and return to a healthy operational state.

Conclusion

Resolving quorum issues in MySQL InnoDB Cluster requires a clear understanding of the current cluster topology and node states. Always prioritize Method 1 (full reboot) unless absolutely necessary to use Method 2 (force quorum), as the latter may risk data inconsistency if misused.